

Ophthalmic Artificial Intelligence From Data Integration to Accountable Clinical Impact: Challenges and Opportunities

Tobias Schlosser^a, Michael Friedrich^b, Richard Vogel^c, Benny Platte^d, Matthias Vodel^c,
Danny Kowerko^b, Marc Ritter^d, Miriam Goldammer^a, Markus Wolfien^{a,e}

^a Dresden University of Technology, Faculty of Medicine Carl Gustav Carus, Center for Medical Informatics,
Fetscherstraße 74, Dresden, 01307, Saxony, Germany

^b Chemnitz University of Technology, Faculty of Computer Science, Junior Professorship of Media Computing,
Straße der Nationen 62, Chemnitz, 09107, Saxony, Germany

^c University of Applied Sciences Mittweida, Faculty of Applied Computer and Biosciences, Professorship of Software Development for
Media and Application Systems, Technikumplatz 17, Mittweida, 09648, Saxony, Germany

^d University of Applied Sciences Mittweida, Faculty of Applied Computer and Biosciences, Professorship of Media Informatics,
Technikumplatz 17, Mittweida, 09648, Saxony, Germany

^e Dresden University of Technology, Center for Scalable Data Analytics and Artificial Intelligence (ScaDS.AI) Dresden/Leipzig,
Strehleener Straße 12, 14, Dresden, 01069, Saxony, Germany

Abstract

Ophthalmology is a leading domain for clinical artificial intelligence (AI) because high-volume imaging, structured measurements, longitudinal follow-up, and screening pathways create unusually clear routes from data to care. We synthesized 427 studies from 6,482 records in PubMed, ACM Digital Library, IEEE Xplore, and Scopus; all included studies addressed ophthalmic imaging data. No lower date limit was set, and studies ranged from January 2010 to April 2026. The evidence shows that ophthalmic AI is moving from image classification toward three clinically important functions: multimodal interpretation, prediction of disease course or treatment need, and clinical workflow integration. Technical performance is no longer the main bottleneck. Accountable clinical value now depends on interoperable data, representative validation, transparent and fair models, privacy and cybersecurity safeguards, regulatory oversight, workflow fit, cost-effectiveness assessment, and post-deployment monitoring. Ophthalmology therefore offers a rigorous model for translating algorithmic performance into clinical benefit.

Keywords: Artificial Intelligence, Ophthalmology, Optical Coherence Tomography, Clinical Translation, Predictive Modeling, Workflow Integration, Governance

1. Introduction

Artificial intelligence (AI) has become increasingly prominent in ophthalmology because the specialty combines high-volume imaging, structured clinical measurements, longitudinal monitoring, and visually interpretable disease phenotypes [4–6]. Optical coherence tomography (OCT) has been especially influential, providing repeatable structural information for diagnosis, segmentation, biomarker extraction, prognosis, treatment-response prediction, and disease monitoring in age-related macular degeneration (AMD), diabetic macular edema (DME), diabetic retinopathy (DR), glaucoma, and retinal vein occlusion (RVO) [7–13]. Ophthalmology therefore offers a concrete test case for a broader digital health question: how can AI move from technical feasibility to measurable and accountable clinical impact?

The broader digital transformation of health care has expanded the expected role of ophthalmic AI. Teleophthalmology, remote monitoring, automated triage, image-quality assessment, multimodal imaging, patient-reported outcomes (PRO), electronic health record data, and large language models (LLMs) are shifting the field from isolated image interpretation toward decision support and care pathway optimization [11, 14–18]. This shift creates opportunities for earlier referral, more consistent

interpretation, individualized monitoring, better use of specialist capacity, and clearer patient communication, while raising the evidentiary threshold for deployment.

Strong benchmark performance can support translation, but it does not by itself establish clinical usefulness. Systems that perform well on curated datasets may fail across imaging devices, institutions, populations, disease spectra, annotation practices, and workflows. Data fragmentation, weak semantic interoperability, limited external validity, biased or unrepresentative training data, insufficient transparency, and unresolved ethical, legal, and regulatory questions should therefore be treated as addressable implementation problems [19–22]. Even when ophthalmic AI reaches expert-level performance in controlled tasks [23], its value depends on whether it improves decisions, outcomes, access, efficiency, or safety under real clinical constraints [24, 25].

This review synthesizes AI in ophthalmology as both a clinical field and a model for AI translation in digital health. It maps the thematic, ethical, methodological, and clinical structure of the literature; identifies where evidence is strongest; and examines conditions for meaningful benefit. For clinicians, computer scientists, health system leaders, regulators, and funders, the

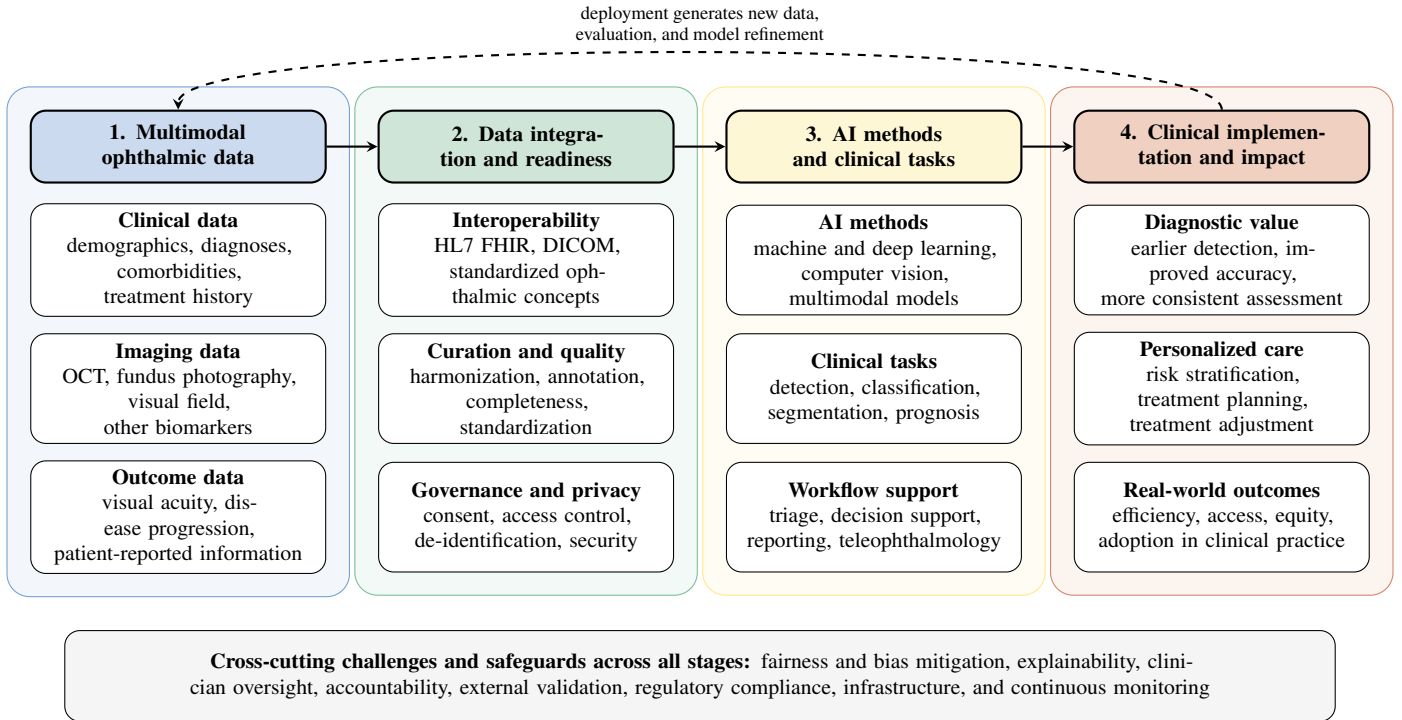


Figure 1: Conceptual overview of artificial intelligence in ophthalmology, tracing the pathway from multimodal ophthalmic data through data integration and AI-enabled clinical tasks to accountable clinical impact. Translation into practice depends on cross-cutting safeguards, including privacy protection, fairness, explainability, regulation, infrastructure, and clinician oversight.

contribution is not an inventory of algorithms, but a translational synthesis of how evidence, data systems, clinical workflows, and governance can be aligned. The review proceeds from search methods and evidence mapping to an evaluation of gaps and a discussion of data foundations, clinical validation, workflow integration, governance, and implementation priorities. Figure 1 provides the conceptual framework, linking multimodal ophthalmic data, data integration and readiness, AI-enabled clinical tasks, and accountable clinical impact.

2. Search strategy and selection criteria

References were identified through structured searches of PubMed, ACM Digital Library, IEEE Xplore, and Scopus, with records captured up to April 30, 2026. No lower limit on publication date was specified a priori. After screening, the 427 included studies ranged from January 2010 to April 2026. Free-text search terms combined AI concepts (“artificial intelligence”, “machine learning”, and “deep learning”) with task terms (“classification”, “detection”, “segmentation”, “monitoring”, “prediction”, and “screening”) and ophthalmic imaging or disease terms (“optical coherence tomography”, “OCT”, “OCTA”, “ophthalm*”, “retina*”, “macula*”, “glaucoma”, “diabetic retinopathy”, and “age-related macular degeneration”). Eligible records addressed AI-based methods in an ophthalmic context, involved OCT, OCTA, or related ophthalmic imaging data, and examined a clinically or translationally relevant task; this imaging requirement defined the review scope. Non-English publications,

duplicates, and records outside the scope of AI in ophthalmology were excluded. The 427-article evidence map is distinct from the broader reference list, which also includes contextual sources.

Three reviewers (TS, MF, and RV) screened titles and abstracts using predefined exclusion criteria. At least two authors were involved in eligibility decisions, and disagreements or borderline cases were resolved by consensus. Articles that passed title and abstract screening underwent full-text assessment. The final reference list was constructed on the basis of eligibility, relevance, methodological importance, and contribution to the thematic, ethical, translational, or evidence mapping synthesis. The review was reported according to PRISMA principles where applicable to this heterogeneous translational review. Article-level mappings, public dataset overview, and extended synthesis are provided in the supplementary material.

3. Evidence synthesis and mapping approach

The literature was synthesized descriptively and qualitatively, complemented by keyword co-occurrence analysis, international co-authorship mapping, public dataset mapping, and evidence gap mapping. For each publication, we extracted article type, ophthalmic condition or clinical context, AI methodology, intended AI role, data-related challenges, implementation barriers, and ethics, governance, or regulatory issues. Extraction and category assignments were reviewed by at least two authors; disagreements were resolved through consensus discussion rather than numerical voting, because the task required categorical interpretation. The supplementary article-level tables provide

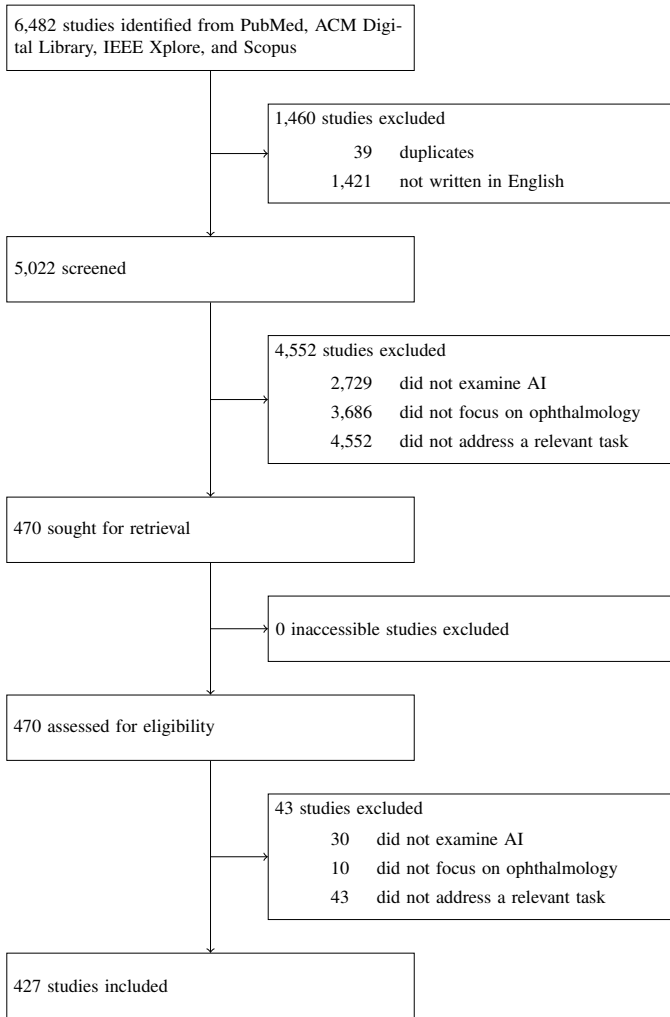


Figure 2: PRISMA-style flow diagram of study identification, screening, eligibility assessment, and inclusion for this systematic review on artificial intelligence in ophthalmology. Records were retrieved from PubMed, ACM Digital Library, IEEE Xplore, and Scopus, and studies were excluded according to predefined criteria at title and abstract screening and full-text review. Overall, 427 studies were included in the final analysis.

the audit trail from extracted fields, contribution summaries, and category assignments to aggregate counts. Evidence gap maps were constructed at article-assignment level and treated as a central interpretive component. General dimensions were categorized as predominantly positive, predominantly negative, or balanced; positive meant enabling the dimension, negative meant creating a barrier or risk, and balanced meant both. For ethics, governance, or regulatory dimensions, “caused” meant AI introduced or amplified an issue, “resolved” meant AI mitigated one, and “both” captured mixed cases such as access gains with bias concerns. Multiple assignments per publication were permitted. These mappings anchored the interpretive synthesis, were not intended to provide pooled diagnostic accuracy estimates, risk-of-bias judgments, or certainty-of-evidence ratings, and provide an explicit bridge between article-level evidence and the roadmap for future research.

4. Ophthalmic AI evidence is expanding from classification to clinical translation

The structured search identified 6,482 analyzable records after database export and harmonization (Fig. 2). The flow diagram reports this harmonized record set rather than raw database-hit totals. After preliminary processing, 1,460 records were excluded, including 39 duplicates and 1,421 non-English publications. The remaining 5,022 records underwent title and abstract screening. At this stage, 4,552 records were excluded on the basis of overlapping thematic filters: 2,729 did not examine AI, 3,686 did not focus on ophthalmology, and 4,552 did not address a task relevant to this review. Because these filters were not mutually exclusive, their counts exceed the total number of excluded records.

A total of 470 studies were sought for retrieval, all of which were accessible and assessed for full-text eligibility. After full-text review, 43 studies were excluded. Again, exclusion categories were not mutually exclusive, with 30 not examining AI, 10 not focusing on ophthalmology, and 43 not addressing a relevant task. Ultimately, 427 studies were included in the synthesis.

Taken together, the literature describes a field with expanding clinical opportunity. Ophthalmic AI has progressed from proof-of-concept classification toward segmentation, quantitative biomarker extraction, prognostic modeling, multimodal integration, screening support, and workflow-oriented applications. Many of these tasks are close to decisions that already structure ophthalmic care: referral, review urgency, treatment timing, monitoring intensity, and follow-up. The next stage is to connect this technical maturity with evidence on implementation, governance, regulation, cybersecurity, fairness, and post-deployment evaluation.

Keyword co-occurrence analysis (Fig. 3) indicated four broad domains: AI methods; OCT-centered retinal imaging and biomarker analysis; disease diagnosis, monitoring, and prognostic modeling; and screening- or workflow-oriented translation. This structure suggests that ophthalmic AI now extends beyond image-level classification toward quantification, multimodal interpretation, longitudinal prediction, and workflow-aware decision support. A similar shift is visible in radiology and pathology, but ophthalmology is distinctive because OCT biomarkers, repeated measurements, screening pathways, and treatment decisions make workflow-linked AI particularly salient.

Bibliometric analysis of geographic distribution and co-authorship (Fig. 4) showed that publication activity and collaboration were concentrated in a relatively small number of countries, especially the United States, China, and India, while parts of Africa and South America were comparatively under-represented. This likely reflects AI research capacity, access to imaging datasets, screening infrastructures, publication volume, and early deployment activity, but these drivers were not formally decomposed. Broader international collaborations across populations, devices, disease presentations, care pathways, and resource settings would strengthen both equity and external validity.

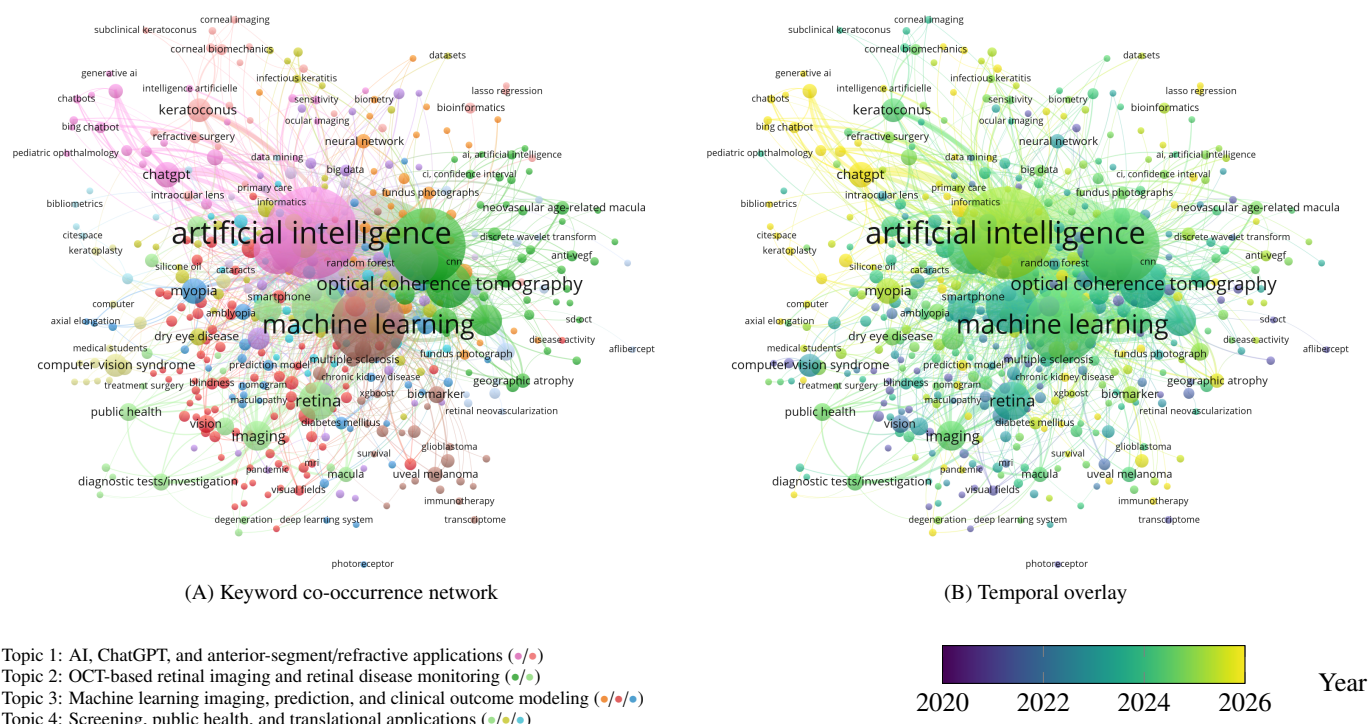


Figure 3: Keyword co-occurrence network of the included literature on artificial intelligence in ophthalmology. (A) Network visualization, in which node size reflects keyword frequency, line thickness indicates co-occurrence strength, and colors indicate data-driven keyword clusters interpreted into thematic areas. Visible labels denote prominent terms, whereas unlabeled nodes represent less frequent connected keywords. (B) Overlay visualization by average publication year, highlighting temporal shifts in research attention. Four major thematic areas can be discerned: core AI methods and computational approaches; OCT-based retinal imaging and biomarker analysis; retinal disease diagnosis, prognosis, and treatment monitoring; and screening, clinical workflow, and emerging translational applications.

The public dataset landscape was similarly concentrated. Reusable benchmark resources were dominated by a relatively small number of datasets for retinal disease classification, segmentation, vascular analysis, and image-quality tasks, including the Kermany dataset, RETOUCH, Duke SD-OCT, OCTA-500, and ROSE-1 [26–28]. These resources have enabled comparison and reproducibility, but they should now be complemented by multi-device, multi-site, longitudinal, and clinically annotated datasets. Benchmark success is necessary for technical progress, but should be the starting point for demonstrating generalizability rather than the endpoint of evaluation. A consolidated overview of identifiable public datasets and benchmark resources is provided in the supplementary material, with dataset-specific references and stable access links where these were identifiable.

5. Evaluation: evidence gaps identify where translation is least mature

We classified the 427 publications across nine general dimensions adapted from recent literature [29]: model performance; technological limitations; real-world impact; data availability, integration, quality, and standardization; infrastructural demands; financial costs; regulatory requirements; cybersecurity risks; and ethical concerns (Fig. 5). This classification yielded 1,094 general-dimension assignments. Each assignment was categorized as predominantly positive, predominantly negative, or balanced according to the publication’s framing rather than proven

effect. The aim was not to estimate efficacy or effect size, but to characterize how the literature distributes attention across technical, translational, infrastructural, and governance domains.

Two observations were central. First, the evidence base was dominated by model performance, technological limitations, real-world impact, and data-related issues. Model performance remained the most prominent dimension overall, with 380 positive and 17 balanced assignments and no negative assignments. Technological limitations accounted for five positive, 171 negative, and 57 balanced assignments, indicating that generalizability limits, image-quality constraints, dataset limitations, and implementation-dependent performance remain central to translation. Real-world impact had 191 positive and seven balanced assignments, but these primarily capture favorable framing of clinical benefit, efficiency, access, or adoption rather than prospective evidence of deployed benefit. Data availability, integration, quality, and standardization accounted for 56 positive, 61 negative, and 72 balanced assignments, emphasizing that data readiness and consistency are determinants of clinical validity, not merely technical prerequisites [30–33]. Because this was a combined category, these counts should not be read as a standalone measure of interoperability or standardization; instead, they show that data availability, integration, quality, and standardization form a shared translational bottleneck. The opportunity is to pair model development with validation, governance, integration, and monitoring from the outset.

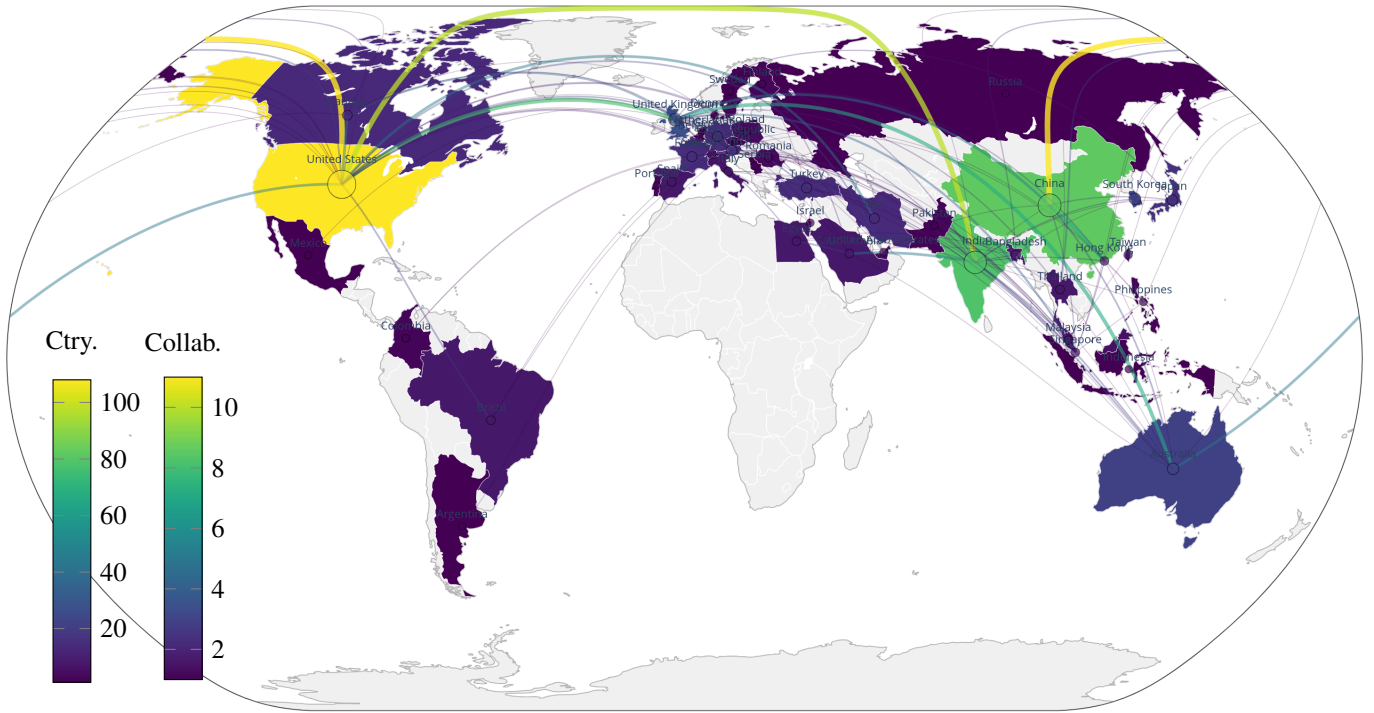


Figure 4: Global distribution of publications and international co-authorship in the included literature on artificial intelligence in ophthalmology. Countries are shaded according to the number of publications, and links indicate international co-authorship relations, with stronger links reflecting more frequent collaboration. The map shows that publication activity and collaboration are concentrated in a limited number of countries, with the strongest publication output and collaboration links centered on the United States, China, and India, while other world regions remain comparatively underrepresented.

Second, implementation-oriented dimensions were underdeveloped and therefore represent priority areas for research. Regulatory requirements were represented by two negative and one balanced assignment, with no positive assignments. Cybersecurity risks accounted for two balanced assignments only. Financial costs were similarly sparse, with six positive, five negative, and five balanced assignments. Explicit ethical concerns appeared as one negative general-dimension assignment, while the ethics-specific map is considered below. This asymmetry suggests that the literature has emphasized feasibility and performance more strongly than deployment conditions, institutional readiness, regulatory pathways, operational security, reimbursement, and post-market responsibilities. US Food and Drug Administration (FDA) authorizations for ophthalmic AI tools show that clinical translation is achievable when intended use, target population, workflow, and evidence requirements are specified [34–36]. They also underscore the need for workflow evaluation, safety monitoring, liability frameworks, and evidence generation after deployment [37].

Assignments indicating negative impact were sparse for real-world impact but prominent for technological limitations, data readiness, and infrastructural demands. This pattern may reflect optimism in clinical framing, but absence of reported negative real-world impact should not be interpreted as absence of harm. It may also indicate limited reporting of failed deployments, workflow burdens, inequitable performance, or adverse downstream effects. For a field moving toward deployment, reports of limited generalizability, poor usability, calibration drift, or oper-

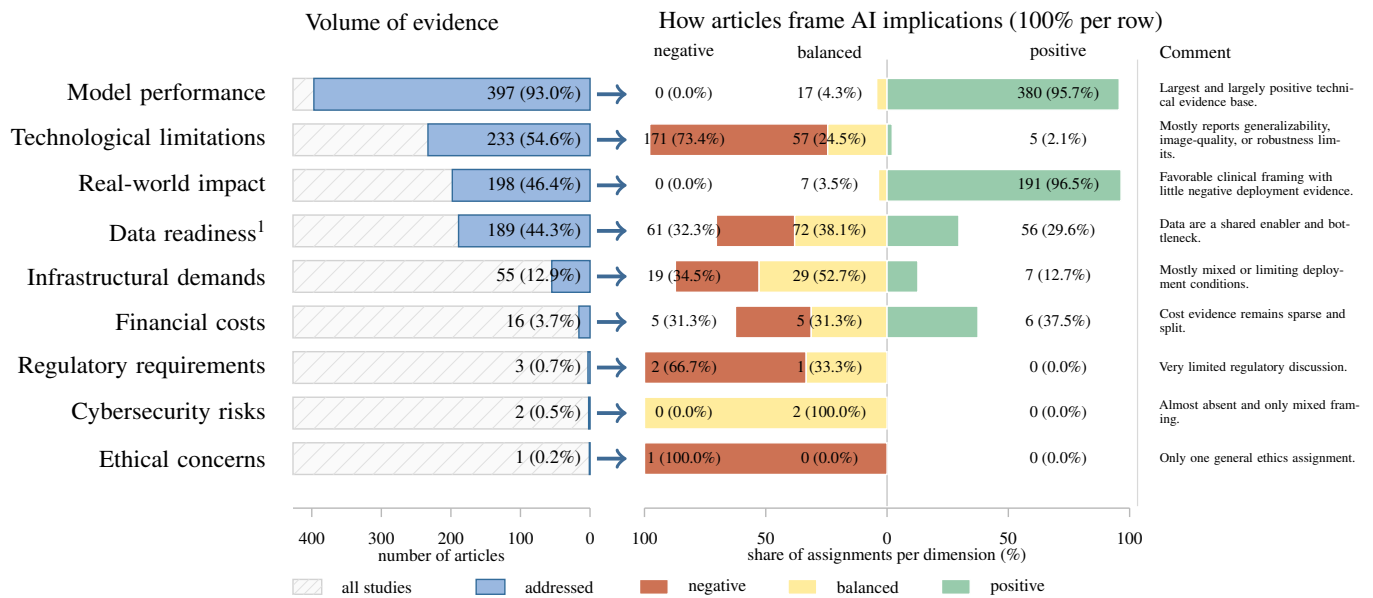
ational failure would be scientifically important. Infrastructural demands showed seven positive, 19 negative, and 29 balanced assignments, indicating recognition that deployment depends on institutional capacity, staffing, procurement, technical integration, maintenance, and workflow adaptation [37–40]. Future work should evaluate clinical effect, implementation conditions, opportunity costs, unintended consequences, and real-world trade-offs with similar rigor.

The following discussion therefore interprets the quantitative evidence-gap patterns rather than introducing a separate evidence base. The high frequency of model-performance, technological-limitation, real-world-impact, and data-readiness assignments, sparse regulatory and cybersecurity assignments, and sparse reporting of negative real-world impact explain the focus on data integration, validation, workflow integration, and governance.

5.1. Ethical, regulatory, and governance enablers

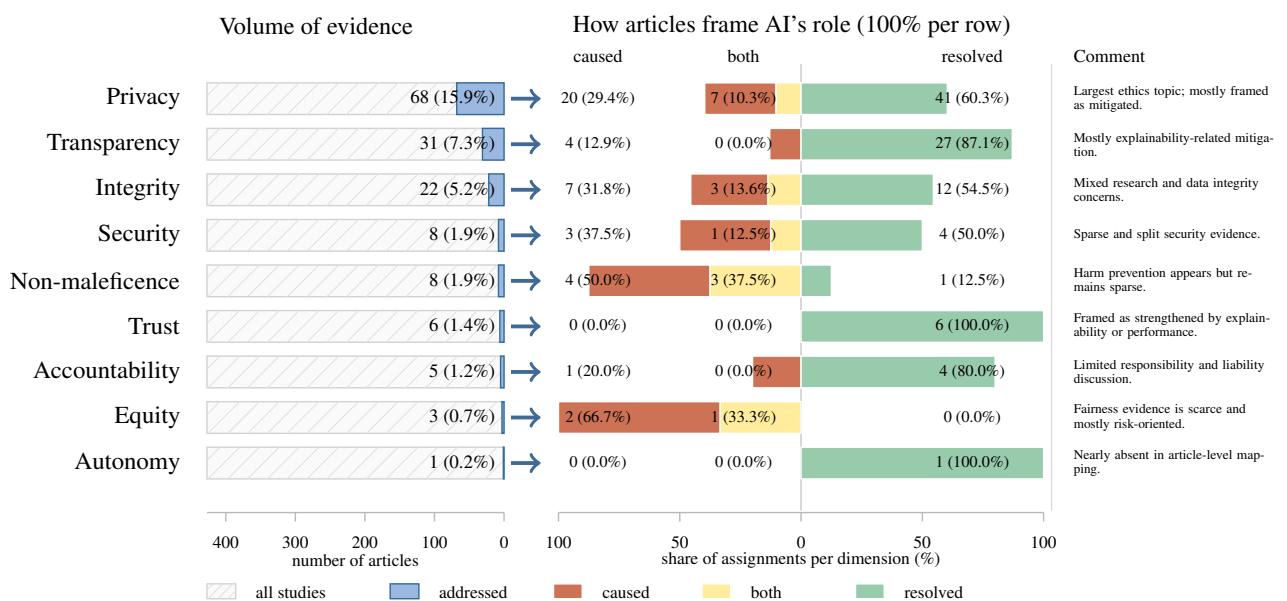
To complement the broader synthesis, we analyzed ethical dimensions adapted from recent literature [41]. Article-level assignments were classified across accountability, autonomy, equity, integrity, non-maleficence, privacy, security, transparency, and trust (Fig. 5) and categorized as caused, resolved, or both caused and resolved. This analysis yielded 152 ethical-dimension assignments. The purpose was to capture the ethical orientation of the literature, not to judge whether individual systems were ethically acceptable.

General dimensions: evidence volume and dominant framing (n=427 articles)



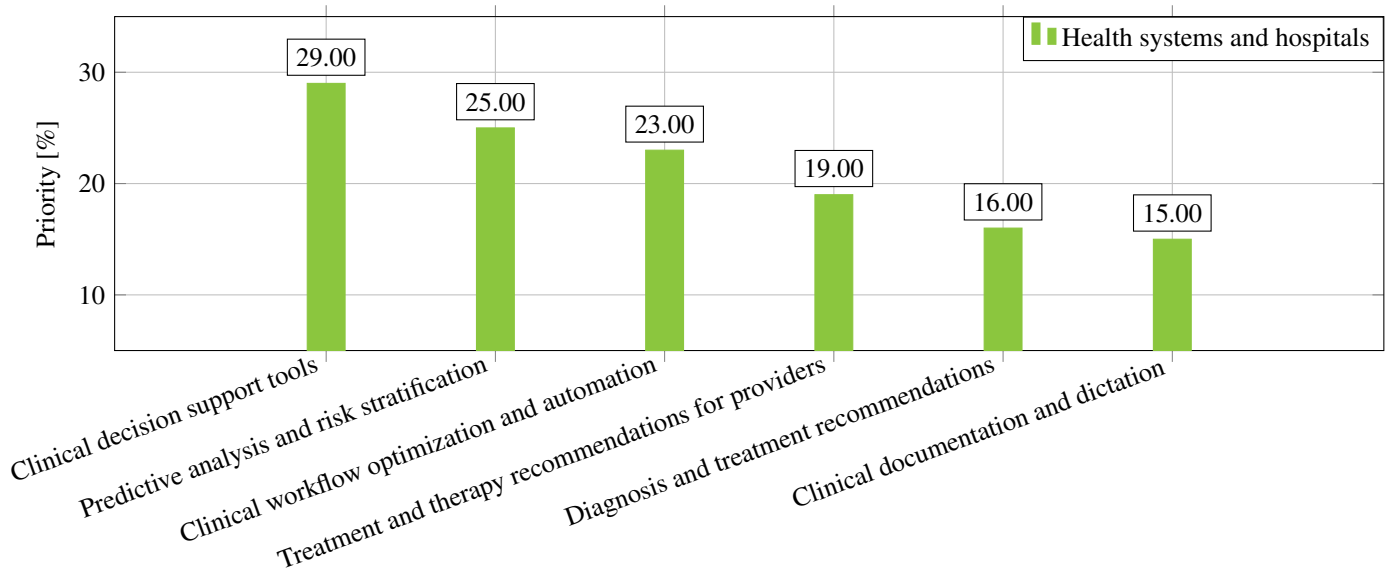
(A) General dimensions

Ethical dimensions: evidence volume and framed role (n=427 articles)

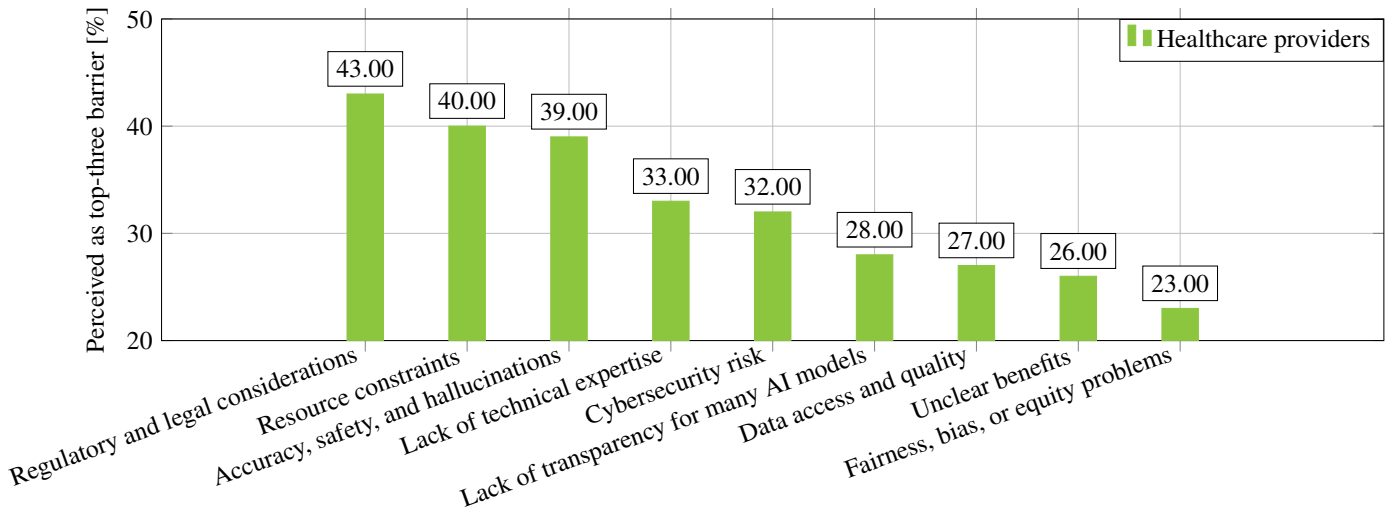


(B) Ethical dimensions

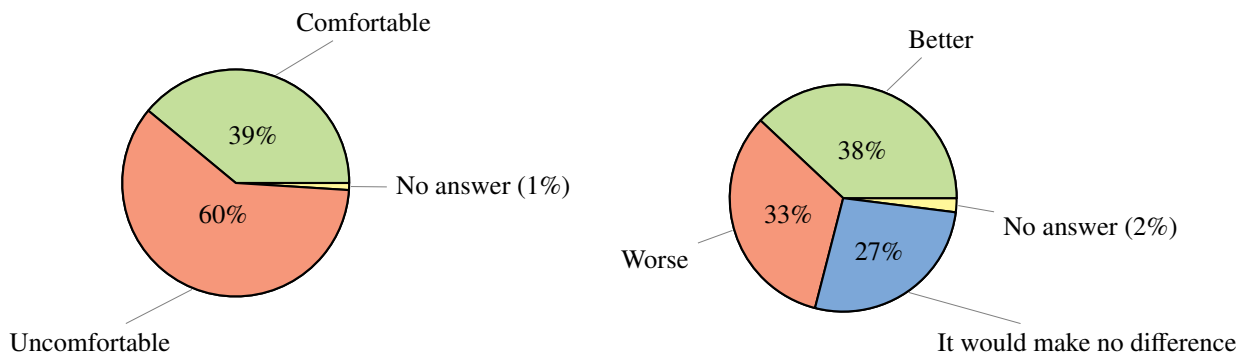
Figure 5: Evidence gap maps summarizing the thematic orientation of the 427 included articles on artificial intelligence in ophthalmology. Each panel combines evidence volume with row-wise framing. **(A)** General dimensions, where the left bars show how many articles addressed each dimension, and the diverging bars show whether the dimension was framed as predominantly positive, negative, or balanced. **(B)** Ethical dimensions, where the left bars show how many articles addressed each ethical dimension, and the diverging bars show whether the issue was framed as caused by AI, resolved by AI, or both. Percentages for diverging-bar categories are calculated within each row. Because individual articles contribute to multiple dimensions, totals across rows and categories exceed the number of included studies.



(A) Priority use cases



(B) Implementation barriers



(C) Public attitudes

Figure 6: Illustrative implementation context for clinical AI from external survey and report sources. **(A)** Use cases aimed at improving care quality, such as clinical decision support tools, are top priorities, with providers also favoring AI applications such as predictive analytics and workflow optimization [1]. **(B)** Despite their significant potential, barriers including regulatory and legal considerations, resource constraints, and model accuracy and safety are impeding broader AI adoption, while the need for governance, transparency, and accountability remains substantial [2]. **(C)** Fewer than half of U.S. adults expect AI to improve patient outcomes in health and medicine, particularly in tasks such as diagnosing diseases and recommending treatments [3]. These data are geographically and institutionally contextual rather than globally representative.

Privacy and transparency were the most prominent ethical themes. Privacy accounted for 20 caused, 41 resolved, and seven caused-and-resolved assignments. It was often framed through privacy-preserving methods such as federated learning, distributed analysis, model-to-data approaches, or synthetic data generation [42–44], but the caused and mixed assignments show that data use, re-identification risk, auditability, and clinical accountability remain active concerns. Transparency accounted for four caused and 27 resolved assignments, with no caused-and-resolved assignment. It was commonly operationalized through saliency maps, class activation maps (CAM), and related visualization strategies [45, 46]. The opportunity is to make image-based decision support more inspectable, reproducible, and measurable, provided explanations are clinically meaningful and linked to uncertainty.

Integrity was the third-largest ethical dimension, with seven caused, 12 resolved, and three caused-and-resolved assignments. Security and non-maleficence each accounted for eight assignments: security had three caused, four resolved, and one caused-and-resolved assignment, whereas non-maleficence had four caused, one resolved, and three caused-and-resolved assignments. Trust accounted for six resolved assignments and no caused or mixed assignments. It was often linked to expert-level or superior diagnostic performance in DR and glaucoma [47, 48]. Performance-based trust remains fragile without external validation, calibration, failure-mode analysis, reproducibility, and clarity about when a system should not be used. Open-source and externally reproducible approaches can strengthen scrutiny but cannot substitute for prospective evidence of clinical usefulness [49, 50]. These counts suggest that privacy, transparency, integrity, security, trust, and non-maleficence are being discussed, but often as technical or procedural mitigations rather than as evaluated governance outcomes.

Accountability was represented by five assignments, equity by three assignments, and autonomy by one assignment. Responsibility should be specified prospectively when model outputs influence triage, diagnosis, referral, or treatment decisions [51, 52]. This is particularly important for tools that alter referral thresholds, prioritize patients for review, or influence treatment intervals. Even where studies discuss equitable access through telemedicine or diverse validation cohorts [51, 53–56], equity is not examined with the same systematic intensity as privacy or transparency. Overall, the ethical synthesis suggests that the literature is strongest on privacy-preserving methods and technical mechanisms for transparency, while the next priority is to examine responsibility, autonomy, equity, non-maleficence, security, and integrity in real care settings.

Ethical, regulatory, and governance considerations should therefore be incorporated into study design, validation, deployment planning, and post-deployment monitoring from the outset.

6. Discussion: accountable value depends on data, workflow, and governance

6.1. Data integration and interoperability remain foundational bottlenecks

Data infrastructure is one of the most consequential opportunities for clinically useful ophthalmic AI. Natural language processing (NLP), multimodal modeling, longitudinal prediction, and workflow support require extensive, high-quality, well-characterized data. Yet the evidence base remains shaped by a limited number of public benchmarks and many retrospective, institution-specific datasets. Reusable benchmarks support comparison and reproducibility, but can narrow the evidentiary basis for generalizability claims. The translational opportunity is to preserve their benefits while adding representative multi-site, multi-device, and longitudinal validation. Synthetic data generation, privacy-preserving analysis, structured data linkage, cross-device harmonization, and integration with electronic health records (EHR) can reduce fragmentation [44, 57, 58], but none removes the need for representative real-world validation.

Interoperability is central because clinically relevant information is distributed across EHR systems, clinical information systems, imaging devices, device-specific analysis software, and local reporting environments. Health Level Seven International Fast Healthcare Interoperability Resources (HL7 FHIR) and DICOM provide foundations for exchange, but syntactic interoperability does not ensure semantic consistency or ophthalmology-specific granularity. Existing terminologies, including the International Classification of Diseases (ICD), the International Classification of Health Interventions (ICHI), and SNOMED CT, remain incomplete for imaging-derived biomarkers, treatment histories, and longitudinal disease states [59, 60]. Closing these gaps would make AI outputs more portable, reproducible, interpretable, and usable in routine decisions. Near-term progress can come from hospital-level inventories of device interfaces, metadata, and data-quality constraints. Mid-term progress requires specialty-level agreement on minimal ophthalmic data elements, shared outcome definitions, and machine-readable ophthalmic concepts across imaging, EHR, and reporting systems. Long-term harmonization should align ophthalmology-specific concepts with HL7 FHIR, DICOM, and clinical terminologies so that registry-scale validation and multi-site workflow integration become feasible.

Data quality is equally decisive. Annotation reliability, completeness, expert adjudication, image quality, device variability, disease-label definitions, and missing clinical context all affect model behavior and usefulness. Structured assessment frameworks and standardized annotation practices indicate methodological maturation [58, 61]. The priority is not simply to accumulate images, but to build ophthalmic data environments that are interoperable, clinically grounded, longitudinally meaningful, and sufficiently representative for safe translation across sites, populations, devices, and health systems. Accordingly, dataset documentation should describe not only the number of images or scans, but also acquisition context, device mix, disease spectrum, label source, missingness, longitudinal follow-up, treatment history, and intended use. Such documentation would

make reported performance easier to interpret and would help reviewers, clinicians, and regulators distinguish portable clinical insight from dataset-specific signal.

6.2. *Clinical translation depends on diagnosis, prediction, and workflow support*

AI-assisted diagnostics remain among the most mature applications in ophthalmology. OCT-based approaches are central because OCT supports detection, classification, segmentation, biomarker extraction, quantitative monitoring, and treatment planning. Deep learning systems have shown strong performance in retinal disease detection and quantification, including geographic atrophy assessment [62], and in selected benchmarks they approach or exceed expert-level performance [47, 48, 63–65]. Teleophthalmology platforms for DR screening demonstrate how technical performance can intersect with public health benefit by increasing access, throughput, and standardization [16, 66, 67]. These examples indicate that ophthalmic AI can extend specialist capacity when the task, intended use, and workflow context are explicit.

The next translational step is clinically situated decision support. Value depends on reliable operation across devices, populations, institutions, image-quality distributions, disease prevalence, and workflow conditions. Multimodal integration, such as combining OCT with fundus photography, OCTA, or structured clinical variables, may align AI outputs more closely with clinical reasoning if harmonization, missing-data handling, validation, interpretability, and workflow design are addressed [68, 69]. Reported productivity gains are encouraging [70], but clinical effect should be judged by patient-relevant and service-relevant outcomes, including timely referral, diagnostic consistency, treatment timing, avoidable visits, workload, safety, equity, and cost-effectiveness. The most valuable systems are likely to be those that make an actionable decision easier, safer, or more consistent, rather than those that merely add a probability score to an already complex clinical encounter.

Predictive modeling is strategically valuable because it shifts attention from recognizing current disease to anticipating clinical trajectories. Models estimating visual acuity (VA), treatment response, recurrence, progression, or retreatment burden could support earlier intervention, individualized follow-up, resource allocation, and shared decision making [12, 71]. Approaches range from regression-based and mathematical models [8, 72] to deep learning systems that stratify progression trajectories [73]. Incorporating OCT biomarkers, treatment history, longitudinal outcomes, and clinical metadata better reflects the temporal structure of ophthalmic care than single-image classification.

Predictive modeling could make ophthalmic care more anticipatory by tailoring monitoring intensity, identifying patients at high risk of progression, and supporting timely treatment adjustment. Evaluation must account for adherence, comorbidity, treatment variation, access to care, clinician decisions, disease severity, and social or behavioral factors that are often incompletely captured. A model may appear useful in a narrow retrospective cohort but lose performance, calibration, or relevance in broader practice. Prediction should therefore be judged

by calibration, uncertainty communication, interpretability, temporal robustness, clinical actionability, and evidence that outputs change decisions that can improve outcomes or efficiency. In practical terms, calibration asks whether predicted risks match observed outcomes in the target setting, temporal robustness asks whether performance remains stable as devices, treatments, and patient mixes change, and actionability asks whether a prediction can trigger an available decision such as review urgency, follow-up interval, treatment adjustment, or patient counseling.

6.3. *Workflow integration makes real-world value testable*

Real-world implementation is where the opportunity of ophthalmic AI becomes clinically testable. Institutional stakeholders prioritize applications that could improve care quality, reduce operational burden, or support resource allocation, including clinical decision support, predictive analytics, and workflow optimization [1]. They also identify barriers, including regulatory and legal uncertainty, limited expertise, cybersecurity concerns, model safety, and unclear return on investment [2]. Public attitudes remain ambivalent, with discomfort often exceeding enthusiasm and fewer than half of surveyed adults expecting improved outcomes from AI-assisted care [3]. This ambivalence defines the evidence and governance conditions needed for patients, clinicians, institutions, and regulators to regard AI as trustworthy, useful, safe, equitable, and governable.

Ophthalmology offers clear examples of deployed AI, including autonomous systems for DR detection [34]. These deployments indicate that translation is possible when the task, intended use, target population, and workflow are specified. Scale-up should fit AI tools into local workflows, reimbursement and regulatory pathways, and clinical information systems, while avoiding additional cognitive or administrative burden. Studies reporting gains in diagnostic consistency or workflow efficiency [6, 73] should be followed by evaluations of patient outcomes, referral appropriateness, treatment timing, workload, access, equity, safety, and cost-effectiveness.

Clinical dashboards can serve as an intermediate step between fragmented data and algorithmic decision support. By assembling OCT biomarkers, visual acuity, treatment history, disease activity, imaging quality, and follow-up intervals, dashboards can make multimodal information usable before an AI model issues predictions. They also expose data-quality problems, support human-in-the-loop review, and make subsequent AI outputs easier to audit.

Implementation science should become a core component of ophthalmic AI research. Progress now depends on demonstrating that AI improves decisions, outcomes, access, equity, safety, or efficiency under real clinical constraints. This requires prospective validation, human-AI interaction studies, workflow evaluation, subgroup assessment, failure-mode monitoring, health-economic analysis, and clinician oversight. Explainability [74], clinician supervision [24, 25], transparent reporting [25, 75], and post-deployment surveillance are therefore core requirements. Evaluations should also report how AI changes the work of clinicians and patients: who reviews outputs, how uncertainty is handled, how disagreements are resolved, and

how system unavailability or erroneous output is managed. Ophthalmic AI should be evaluated as a clinical intervention embedded in a sociotechnical system, not simply as a model applied to images.

6.4. From technical feasibility to accountable clinical value

The evidence reviewed here indicates that ophthalmic AI now encompasses diagnosis, segmentation, quantitative biomarker extraction, prediction, screening, teleophthalmology, and workflow support. The field is increasingly defined not by whether algorithms can recognize disease from images, but by whether systems generate interpretable outputs, generalize across devices and populations, support longitudinal care, and fit safely into workflows. Ophthalmology therefore illustrates the broader transition from technical feasibility to accountable clinical value. The strongest evidence concerns model performance, image interpretation, and data-related enablers. The next gains are likely to come from linking these strengths to regulation, cybersecurity, fairness, implementation readiness, and post-deployment evaluation.

Accountable clinical value requires more than retrospective benchmark performance. It requires representative data, semantic interoperability, transparent reporting, external and prospective validation, subgroup and fairness assessment, privacy-preserving governance, workflow evaluation, and post-deployment monitoring. These requirements are the conditions that allow technically successful systems to be trusted, adopted, and sustained in routine care. They also provide a constructive research agenda. Each requirement can be translated into measurable study design choices, reporting standards, procurement criteria, or regulatory evidence.

Adjacent image-rich specialties such as radiology and pathology show that AI implementation is a service-design problem as much as a model-development problem. Ophthalmology can learn from radiology's experience with intended-use specification, image-archive and reporting integration, audit trails, and prospective workflow evaluation, while pathology highlights the importance of standardization, annotation quality, and human oversight when AI output changes diagnostic work. Ophthalmology can in turn contribute lessons on longitudinal imaging, screening pathways, treatment-interval decisions, and patient-facing monitoring. Multimodal and agentic AI systems may eventually coordinate imaging, structured clinical data, free-text notes, patient-reported information, scheduling, and follow-up recommendations. Their value will depend on foundational conditions, encompassing interoperable data, explicit intended use, reliable validation, clinician oversight, audit trails, and mechanisms for detecting drift or harm.

The following roadmap summarizes practical priorities for moving ophthalmic AI from technical feasibility toward accountable clinical value.

Roadmap for clinically valuable and accountable ophthalmic AI

1. **Data and protocol foundations.** Report provenance, device context, longitudinal structure, ophthalmic concept definitions, and reference-standard procedures.
2. **Clinical validation and utility.** Specify intended use, target population, care setting, and decision threshold, then test effects on decisions, outcomes, workload, access, safety, equity, and costs.
3. **Workflow integration.** Design outputs around actions such as referral, monitoring, follow-up interval, treatment timing, or counseling, with clinician oversight.
4. **Fairness and governance.** Evaluate subgroup and intersectional performance, quantify uncertainty, and embed privacy, cybersecurity, auditability, regulatory evidence, and drift monitoring.
5. **Next-generation decision support.** Develop multimodal, language-model, or agentic functions only when intended use, human responsibility, uncertainty, escalation pathways, and post-deployment surveillance are explicit.

7. Strengths and limitations

This review combines a structured search with thematic analysis, keyword co-occurrence mapping, international co-authorship analysis, public dataset mapping, and evidence gap mapping. This approach provides a broader perspective than a narrative review focused only on selected applications or model performance. The supplementary material provides article-level mappings and dataset documentation to support transparency and traceability.

Several limitations should be acknowledged. First, no lower limit on publication date was specified a priori, but the article-level evidence map ultimately included studies from January 2010 to April 2026. This window reflects eligible literature captured by the search and screening process, not a deliberately bounded review beginning in 2010; studies after the April 30, 2026 search cutoff may alter the balance of evidence. Second, the review was imaging-dependent because eligibility required OCT, OCTA, or related ophthalmic imaging data, and therefore does not fully capture non-imaging ophthalmic AI such as administrative NLP or operational analytics. Third, only English-language publications were included, which may have excluded relevant work in a globally distributed field. Fourth, heterogeneity in article type, clinical indication, AI task, imaging modality, dataset source, validation strategy, and outcome reporting limited comparability and precluded meta-analysis. Fifth, evidence gap maps summarize article-level orientations rather than standardized effect sizes, diagnostic accuracy estimates, risk-of-bias assessments, or certainty ratings. Finally,

many studies provided limited information on sex, gender, race, ethnicity, socioeconomic position, or other dimensions of representativeness.

8. Conclusion

AI in ophthalmology has matured into a diverse field with clear potential in diagnosis, segmentation, biomarker extraction, prediction, screening, teleophthalmology, and workflow support. Technical progress, especially in OCT-based applications, creates a realistic opportunity to improve access, consistency, monitoring, personalization, and operational efficiency. Whether this opportunity becomes clinical value now depends less on algorithms alone than on data readiness, semantic interoperability, representative validation, fairness, privacy, cybersecurity, workflow integration, regulatory readiness, and sustained clinical oversight. This standard is plausible in ophthalmology because the field combines repeatable imaging, quantitative biomarkers, longitudinal follow-up, screening pathways, and actionable monitoring decisions.

The next phase should be opportunity-driven but evidence-disciplined. Future work should prioritize interoperable data, validation beyond benchmarks, transparent reporting, subgroup and fairness assessment, privacy-preserving governance, prospective workflow evaluation, cost-effectiveness assessment, and post-deployment monitoring. These priorities are enablers of clinical benefit: they help useful systems scale, support clinicians, and keep AI-supported care accountable. The central question is not simply whether AI can detect ophthalmic disease, but whether it can be made robust, explainable, equitable, governable, and clinically useful in real care settings. If this standard is met, ophthalmology could become one of the clearest examples of digital health moving from algorithmic performance to accountable clinical benefit.

Contributors

T.S., R.V., and M.W. conceptualized the review. T.S. and M.F. developed the methodology. T.S. and R.V. designed and implemented the software and data processing workflow, provided resources, and curated the article-level data. T.S. and M.F. performed the validation and formal analysis. T.S., M.F., and R.V. screened records, assessed eligibility, extracted data, and interpreted the findings. T.S. created the original visualizations, and B.P. contributed visualization concepts and figure feedback, including the evidence-map presentation. B.P., M.V., D.K., M.R., M.G., and M.W. provided substantive manuscript feedback, interpretation guidance, and critical review of the structure, methods, results, and discussion. T.S., M.F., R.V., and M.W. drafted the original manuscript, and all authors critically reviewed the manuscript. T.S., M.F., and R.V. had full access to the article-level data underlying the review and verified the data. All authors reviewed and approved the final version of the manuscript and take responsibility for the manuscript content and the decision to submit.

Declaration of interests

We declare no competing interests.

Role of the funding source

No specific funding was received for this review. No funder had any role in the design of the review, in the collection, analysis, or interpretation of the literature, in the writing of the manuscript, or in the decision to submit the manuscript for publication.

Data sharing

No individual participant data were collected or analyzed for this review. The article-level extraction tables, evidence gap mappings, and dataset overview underlying the review are provided in the supplementary material. Additional analysis code or extraction files can be made available by the corresponding author upon reasonable request.

Declaration of AI usage

OpenAI Codex (GPT-5.5) was used to support language editing, consistency checks, and preparation of the supplementary material. AI-assisted tools were not used as autonomous evidence sources or final arbiters of screening, eligibility, extraction, or evidence-gap coding decisions. All AI-assisted output was reviewed, edited, and approved by the authors, who remain responsible for the manuscript content and decision to submit.

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